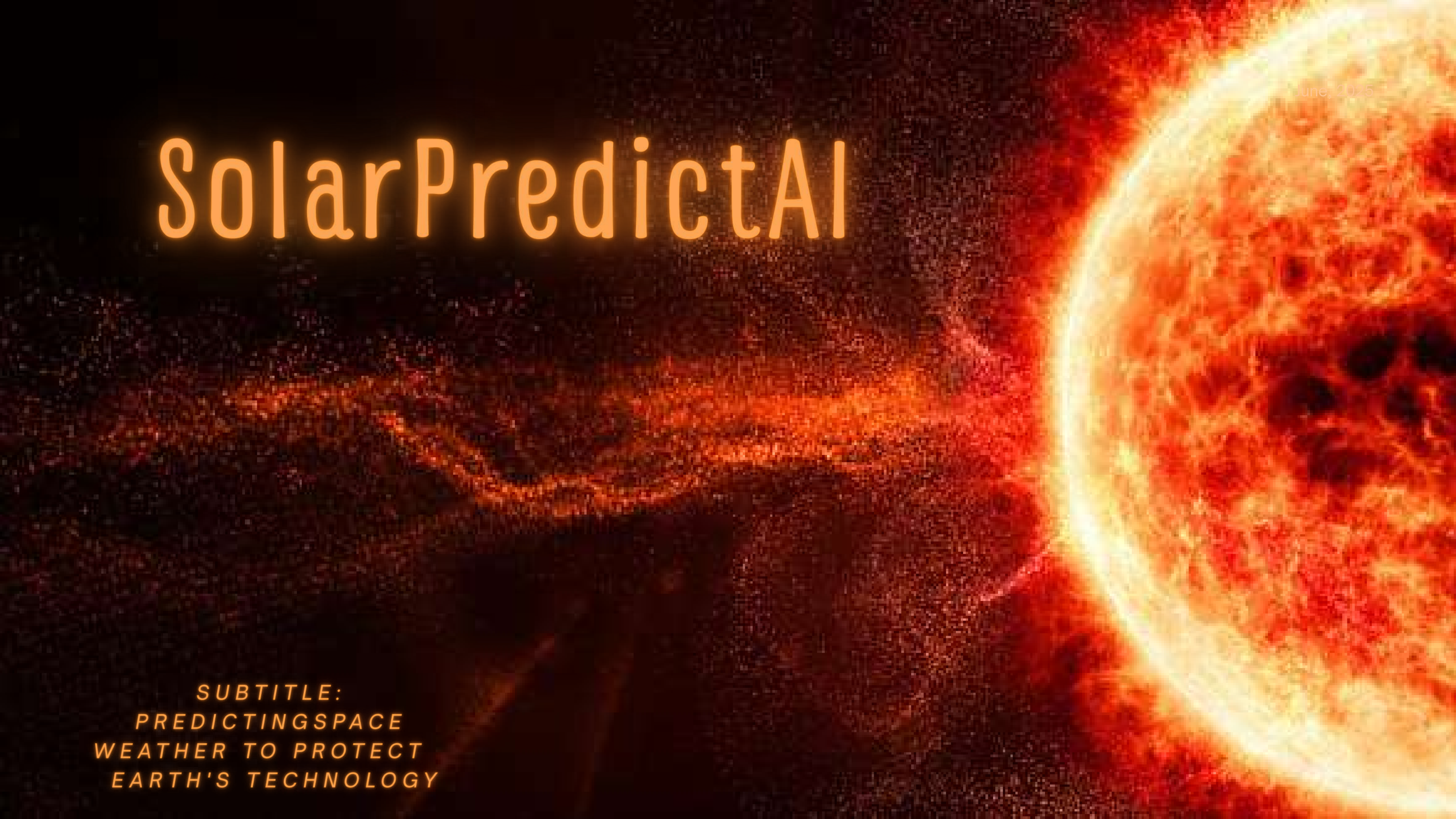
SolarPredictAI



SUBTITLE:
PREDICTINGSPACE
WEATHER TO PROTECT
EARTH'S TECHNOLOGY

The Problem

Our tech-dependent world is dangerously exposed to space.
Solar flares and CMEs can disable satellites, causing $X billion in losses.
GPS and radio blackouts disrupt aviation and global logistics.
Geomagnetic storms can destroy power grids
Current forecasting methods are either too general, lack precision, or arrive too late to act

The Solution

What We Do?

We have built an AI system that doesn't just predict a flare, but models its entire journey and forecasts its impact on specific regions of Earth.

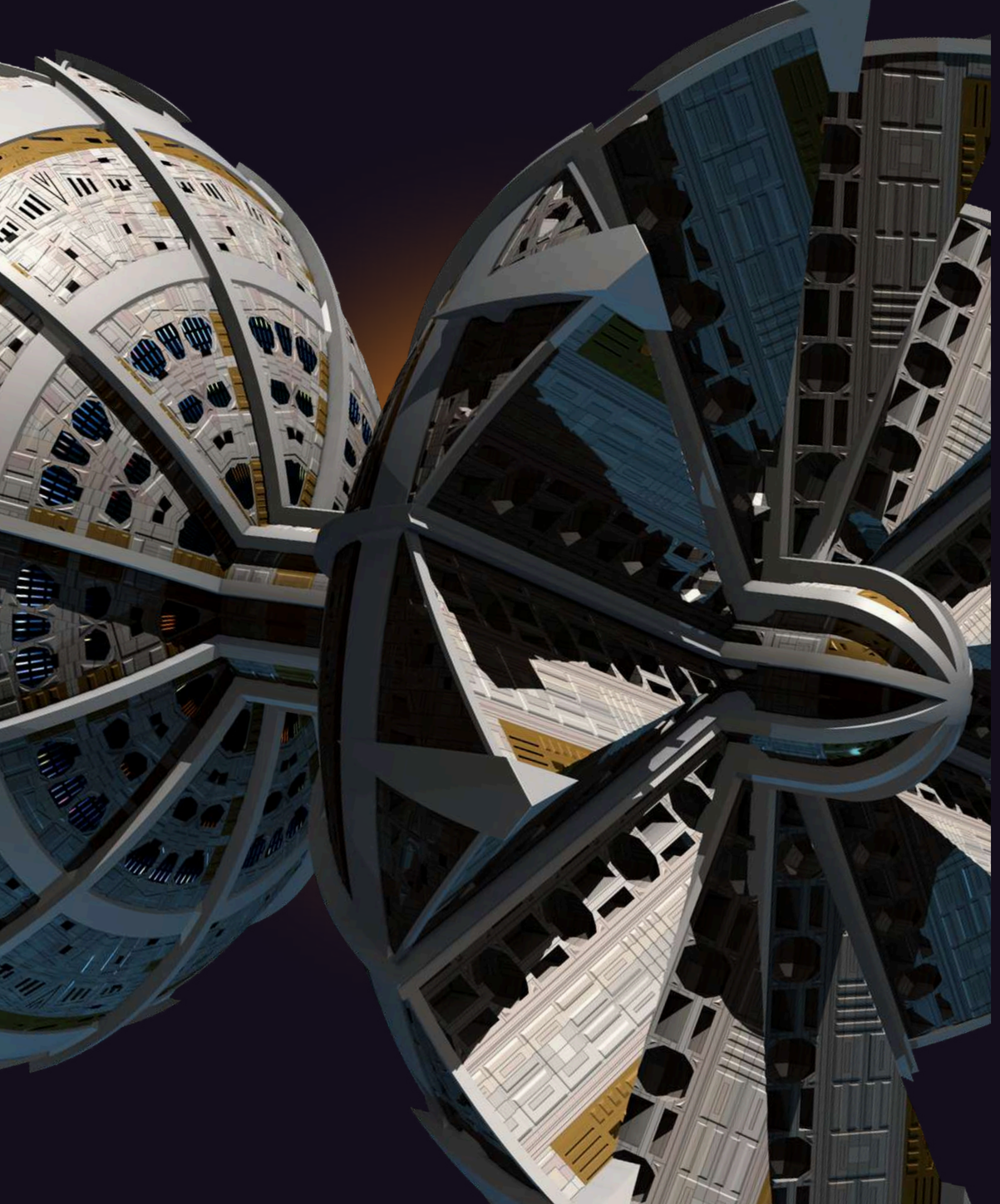
Our Unique Approach:

Uniting three critical stages in one AI:

1. Predicting the probability of a solar flare.
2. Modeling the trajectory of the resulting CME.
3. Localizing the geomagnetic impact on a map.

What's new?

The geosphere is the part of Earth that includes all the rocks, minerals, landforms, and the ground beneath our feet. It includes mountains, valleys, and all types of soil. Think of the geosphere as the solid, non-living part of Earth.



How It Works?

1. Input Data: Solar images (SDO), magnetograms, historical flare data, geomagnetic indices.

2. Model 1 (CNN + LSTM): Image and time series analysis → flare probability forecast.

3. Model 2 (PINN + Physics): CME trajectory and Earth's magnetosphere interaction calculation.

4. Output: Interactive risk map + recommendations

We've developed a simple and intuitive interface

THE USER CONFIGURES MISSION PARAMETERS AND SELECTS A REGION

The Product

Interface Features:

- Risk Map: An interactive world map color-coded from green to red, highlighting regions at risk of power grid disruptions and communication failures.

- Timeline: Forecasts available for 24, 48, and 72 hours.

- Actionable Recommendations: Automated alerts for satellite operators ("enter safe mode") and energy companies ("secure grid reserves").

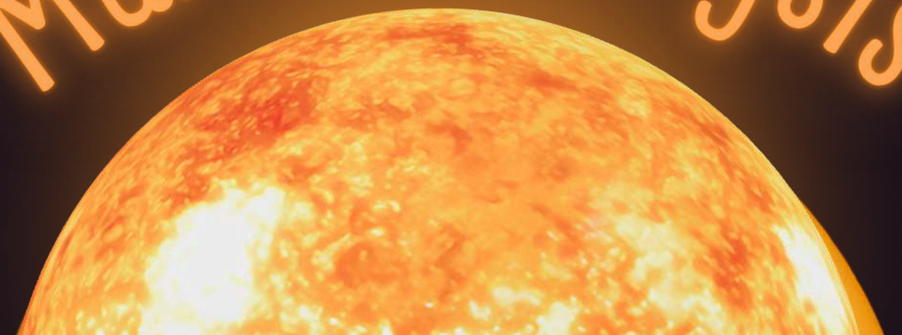
A growing market with a critical need.

TAM (Total Addressable Market): The global space weather and infrastructure protection market. Estimated at $X Billion (Source: NOAA, SIA, MarketsandMarkets).

SAM (Serviceable Available Market): Satellite insurance, energy companies, and government agencies (NASA, ESA, NOAA, national emergency services).

SOM (Serviceable Obtainable Market): Initial pilot projects with national power grid operators and satellite communication firms within the first 3 years.

Market Analysis



Others see a cloud; we see the strike.

Competitors (SWPC NOAA, Met Offices) SolarPredictAI
Forecasts a general level of activity. Forecasts the geographic location of the impact.
Lag time of tens of minutes to hours. Prediction based on early indicators on the Sun (AI-driven).
Based on empirical rules and thresholds. Based on data-driven physical models (PINN) .
No specific, automated advice. Provides actionable recommendations for specific industries.

Business Model

Revenue Streams:

B2B Subscriptions: Monthly fees for API access and web dashboard
· Enterprise Integration: One-time implementation fee + annual maintenance
· Government Contracts: Long-term agreements with space agencies and emergency services

Target Clients:

Satellite operators
· Power grid companies
· Insurance firms
· Aviation and logistics
· Government agencies

Value Proposition:

Protect satellites (worth $10M+ each)
· Prevent power grid failures
· Reduce insurance premiums
· Ensure GPS and communication reliability

A Comprehensive Approach

Analyzes real-time solar images

Shows which Earth regions will be at risk

Predicts the probability of solar flares

Provides recommendations for infrastructure protection

The Team

Data Scientist: [Nurridin]. Expert in handling astronomical datasets (NASA SDO, SOHO). Skilled in cleaning and preparing large-scale data.

ML Engineer: [Ayazhan]. Specialist in Computer Vision (CNN) and time-series analysis (LSTM). Experience implementing PINNs for industrial applications.

Visualization & Analytics): [Aiaru]. Data visualization expert. Creates intuitive interfaces and interactive maps to explain complex scientific processes.

Thank You